between the predictions and the actual observed motion of such objects is spectacular. This success gave Newton's theory unequivocal support until the early part of the twentieth century. Einstein's discovery of special relativity, however, raised what proved to be an insurmountable obstacle for Newton's theory.

The Incompatibility of Newtonian Gravity and Special Relativity

A central feature of special relativity is the absolute speed barrier set by light. It is important to realize that this limit applies not only to material objects but also to signals and influences of any kind. There is simply no way to communicate information or a disturbance from one place to another at faster than light speed. Of course, the world is full of ways for transmitting disturbances at slower than the speed of light. Your speech and all other sounds, for example, are carried by vibrations that travel at about 700 miles per hour through air, a feeble rate compared with light's 670 million miles per hour. This speed difference becomes obvious when you watch a baseball game, for instance, from seats that are far from home plate. When a batter hits the ball, the sound reaches you moments after you see the ball being hit. A similar thing happens in a thunderstorm. Although lightning and thunder are produced simultaneously, you see the lightning before hearing the thunder. Again, this reflects the substantial speed difference between light and sound. The success of special relativity informs us that the reverse situation, in which some signal reaches us before the light it emits, is just not possible. Nothing outruns photons.

Here's the rub. In Newton's theory of gravity, one body exerts a gravitational pull on another with a strength determined solely by the mass of the objects involved and the magnitude of their separation. The strength has nothing to do with how long the objects have been in each other's presence. This means that if their mass or their separation should change, the objects will, according to Newton, immediately feel a change in their mutual gravitational attraction. For instance, Newton's theory of gravity claims that if the sun were suddenly to explode, the earth-some 93 million miles away—would instantaneously suffer a departure from its usual elliptical orbit. Even though it would take light from the explosion eight minutes to travel from the sun to the earth, in Newton's theory knowledge that the sun had exploded would be instantaneously transmitted to the earth